PARISHRAM 2.0

CLASS 12TH 2026

Electric Charges and Fields

PHYSICS LECTURE-7

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Topics to be covered:

1. Electric Flux
2. Gauss's Law
3. Electric Field due to a Point charge (By Gauss's Law)
4. Electric Field due to an infinite linear charge distribution

Recap

1. Electric Field due to charged Ring on the axis
2. Electric Field due to a Uniformly charged straight rod
3. Electric Lines of Force and its characteristics

Electric Flux

- The strength of the electric field is measured by this physical quantity (also called flux density).
- The electric flux can be understood graphically by the total number of electric field lines passing through an area.
- For an area element $d\vec{A}$, placed in an electric field $\vec{E}$ as shown, electric flux is defined as $d\phi = \vec{E} \cdot d\vec{A}$.

$$\phi_E = \vec{E} \cdot \vec{A}$$

$$\Rightarrow \phi_E = EA\cos\theta$$

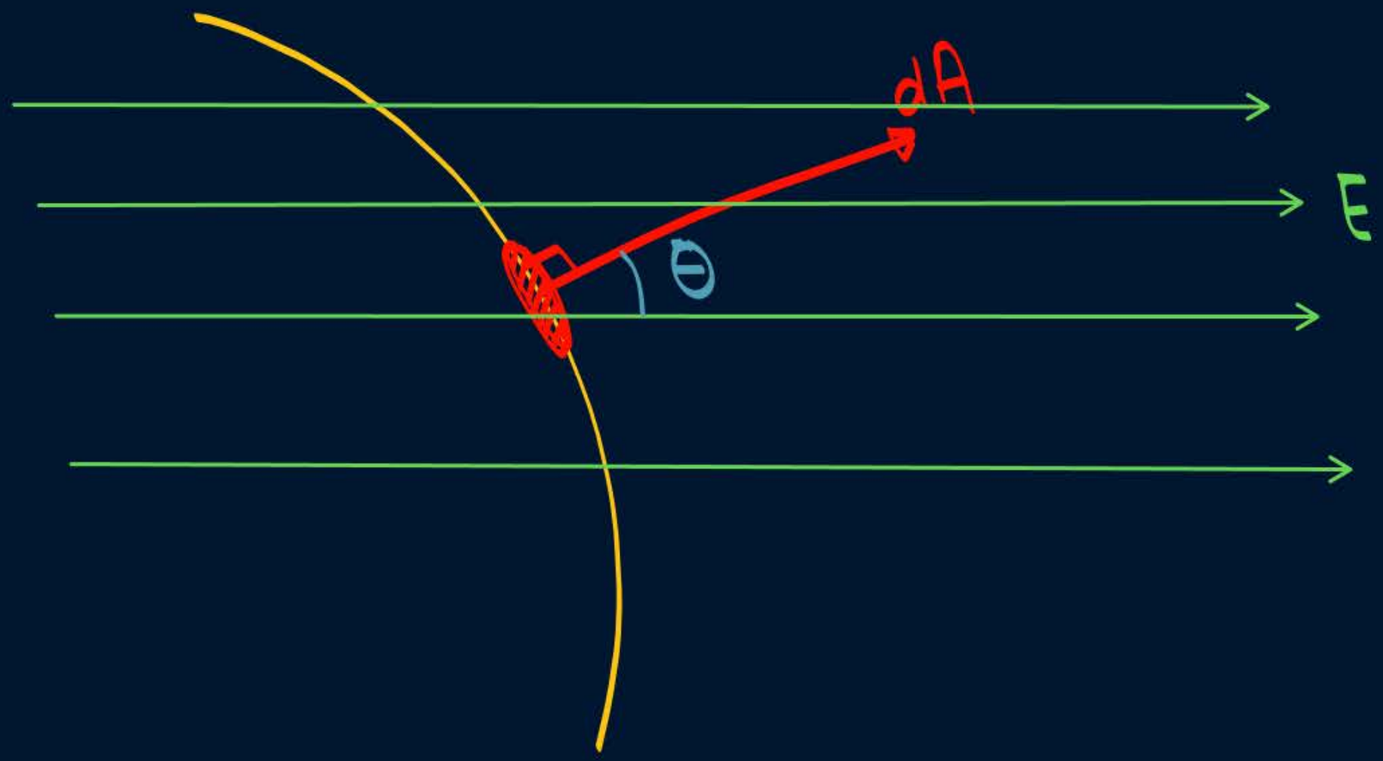
$$\phi_E = \frac{Q_{enclosed}}{\epsilon_0}$$

Total charge enclosed $= q_1 + q_2 + q_3 + q_4$.



Electric Flux

Area vector is always perpendicular to the surface.



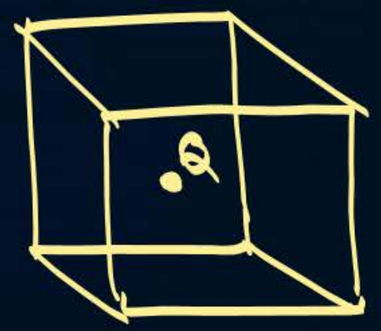
Electric flux through the small area element $d\vec{A}$ = $d\phi$.

$$d\phi = \vec{E} \cdot d\vec{A}$$

$$\phi_E = \int d\phi = \int E \cdot dA$$

$$\Rightarrow \boxed{\phi_E = E \int dA}$$

Que :-



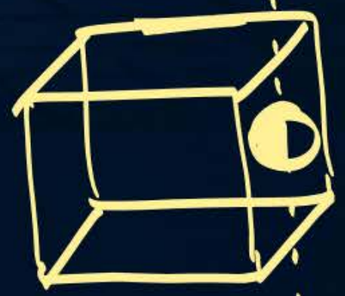
Find the Electric flux through the cube ?

A) $\frac{2Q}{\varepsilon_0}$

B) $\frac{Q}{\varepsilon_0}$

C) $\frac{4Q}{\varepsilon_0}$

D) $\frac{Q}{2\varepsilon_0}$



$\phi_\epsilon = ?$

A) $\frac{Q}{\epsilon_0}$

C) ~~$\frac{Q}{2\epsilon_0}$~~

B) $\frac{Q}{4\epsilon_0}$

D) $\frac{Q}{6\epsilon_0}$

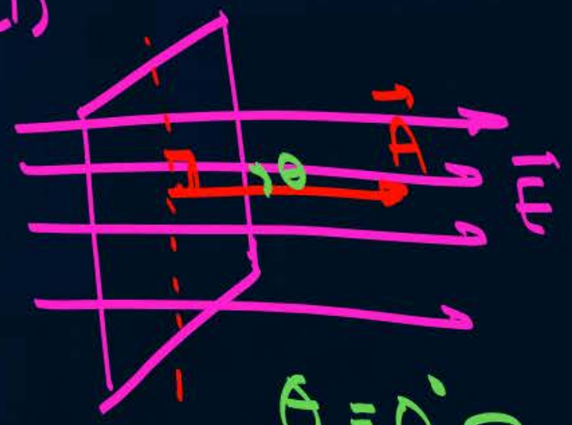
$Q_{enclosed} = \frac{Q}{2}$

$\phi_\epsilon = \frac{Q/2}{\epsilon_0} = \boxed{\frac{Q}{2\epsilon_0}}$

$$\phi_E = \vec{E} \cdot \vec{A} \Rightarrow \boxed{\phi_E = EA\cos\theta}$$

Special Cases :-

(i)

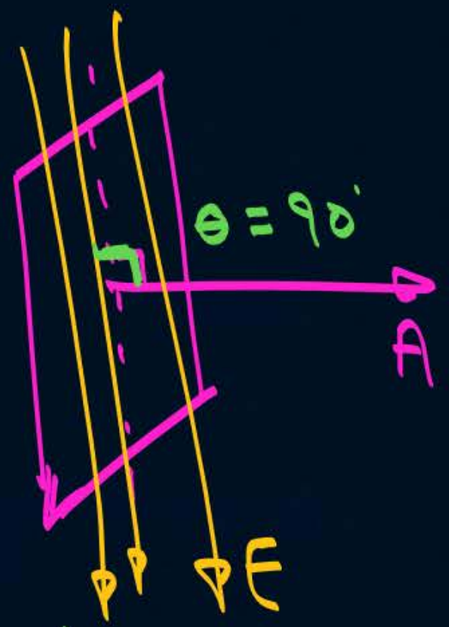


$\theta = 0^\circ$

$\phi_E = EA\cos\theta$

$\boxed{\phi_E = EA}$

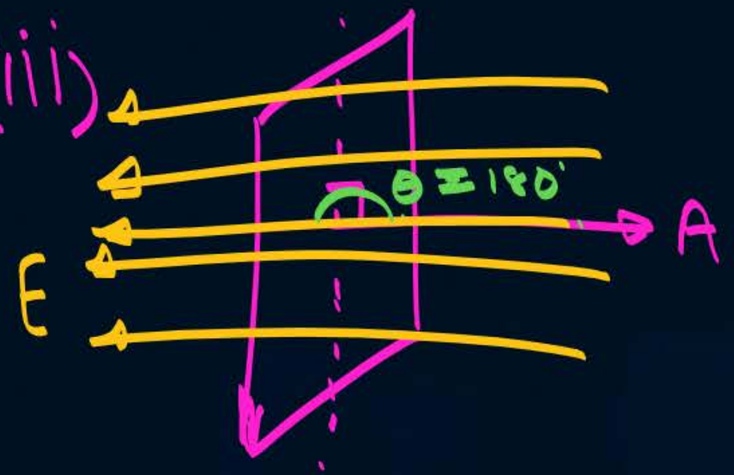
(ii)



$\phi_E = EA\cos 90^\circ$

$\boxed{\phi_E = 0}$

(iii)



$\phi_E = EA\cos(180^\circ)$

$\boxed{\phi_E = -EA}$

QUESTION

A point charge of $5\mu C$ is placed at the center of a spherical surface of radius 0.1 m . The electric flux through the sphere is:
$(\varepsilon_0 = 8.85 \times 10^{-12} C^2/Nm^2)$

$\phi = \frac{q}{\varepsilon_0} = \frac{5 \times 10^{-6}}{8.85 \times 10^{-12}}$

A ✓ $5.65 \times 10^5 Nm^2/C$

B $4.52 \times 10^5 Nm^2/C$

C $2.82 \times 10^5 Nm^2/C$

D $3.39 \times 10^5 Nm^2/C$



Electric Flux DPP ✓

THANK YOU

